

Math 344, S19
Quiz 8

Name: Answer

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Are these pairs of vectors **orthogonal** to each other or only **linearly independent**?

(a) $\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. Linear independent

$$\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 2 \neq 0$$

(b) $\begin{bmatrix} -0.8 \\ -0.6 \end{bmatrix}, \begin{bmatrix} -0.3 \\ 0.4 \end{bmatrix}$. orthogena

$$\begin{bmatrix} -0.8 \\ -0.6 \end{bmatrix} \cdot \begin{bmatrix} -0.3 \\ 0.4 \end{bmatrix} = 0.24 - 0.24 = 0$$

(c) $\begin{bmatrix} \cos \theta \\ \sin \theta \\ 1 \end{bmatrix}, \begin{bmatrix} -\cos \theta \\ -\sin \theta \\ 1 \end{bmatrix}$. orthogena

$$\begin{bmatrix} \cos \theta \\ \sin \theta \\ 1 \end{bmatrix} \cdot \begin{bmatrix} -\cos \theta \\ -\sin \theta \\ 1 \end{bmatrix} = -\cos^2 \theta - \sin^2 \theta + 1 = 0$$

Problem 2. (5 points) Given two vectors $\mathbf{a}_1 = (1, 1, 1)$ and $\mathbf{a}_2 = (0, 1, 2)$.

(a) Find the projection of $\mathbf{b} = (4, 2, 6)$ onto the subspace of $\mathbf{R}^3$ spanned by $\mathbf{a}_1$ and $\mathbf{a}_2$.

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \Rightarrow A^T A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}$$

$$\mathbf{b} = (4, 2, 6) \Rightarrow A^T \mathbf{b} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \\ 6 \end{bmatrix} = \begin{bmatrix} 12 \\ 14 \end{bmatrix}$$

$$\text{solve } A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$$

$$\begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix} = \begin{bmatrix} 12 \\ 14 \end{bmatrix}$$

$$\Rightarrow \hat{x}_1 = 3 \quad \hat{x}_2 = 1$$

$$\text{Thus } \mathbf{p} = A \hat{\mathbf{x}} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}$$

(b) Find the projection matrix $\mathbf{P}$ for the subspace of $\mathbf{R}^3$ spanned by $\mathbf{a}_1$ and $\mathbf{a}_2$.

$$(A^T A)^{-1} = \frac{1}{6} \begin{bmatrix} 5 & -3 \\ -3 & 3 \end{bmatrix}$$

$$\mathbf{P} = A(A^T A)^{-1} A^T = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \frac{1}{6} \begin{bmatrix} 5 & -3 \\ -3 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

$$= \frac{1}{6} \begin{bmatrix} 5 & 2 & -1 \\ 2 & 2 & 2 \\ -1 & 2 & 5 \end{bmatrix}$$